Project: Analyze Customer Behavior

We are going to analyze the entire customer lifecycle: acquisition, conversion, customer activity, retention and churn.

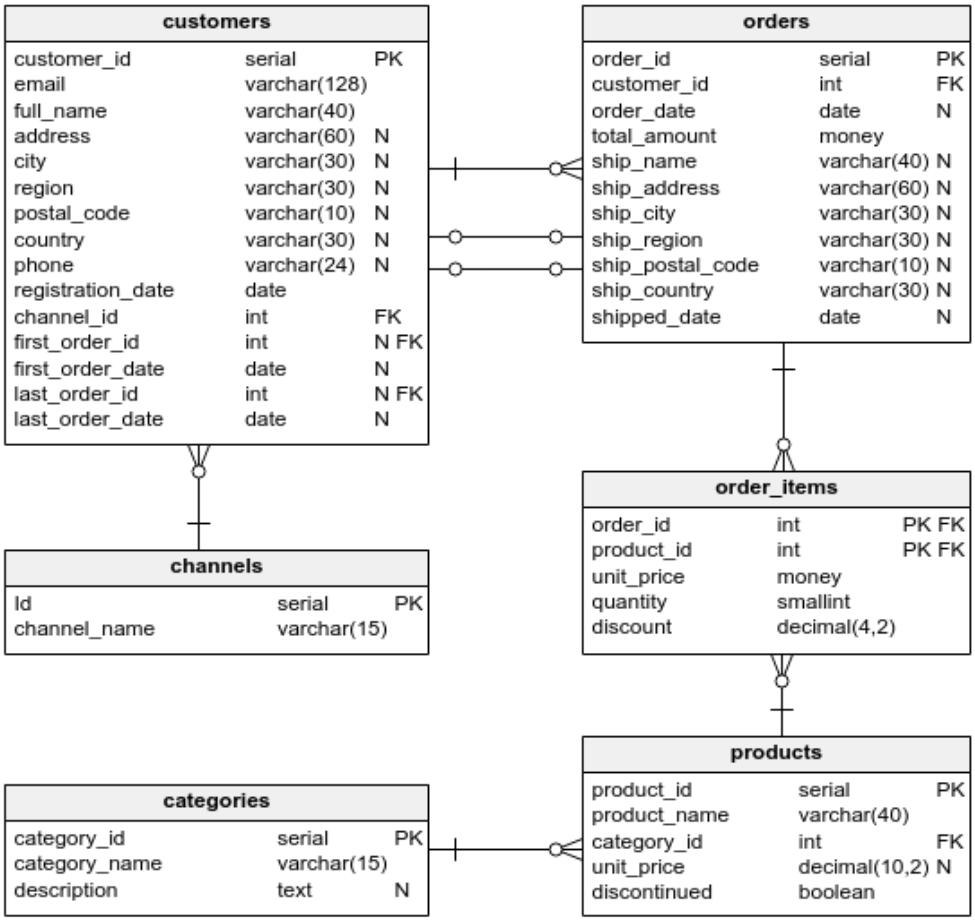
acquisition is the stage when the customers learn more about the company's offerings from visits to the website, in an online business, this stage ends with customer registration on the website.

Conversion is the stage where customers purchase a product or service.

Retention. Now that a new customer has been acquired, the focus is to help the customer derive satisfaction and value from the product and services. In this stage, you can measure **customer activity**: how much they purchase, how often they use your product, etc.

The final stage in the customer lifecycle is **customer churn**. Inevitably, customers will stop using your product

Data Model



Explanation about relationship

- Customers and Orders: A customer can have many orders, but an order belongs to only one customer. This is represented by the foreign key customer_id in the orders table. [One-to-Many Relationships]
- Orders and Order Items: An order can have many order items, but an order item belongs to only one order. This is represented by the foreign key order_id in the order_items table. [One-to-Many Relationships]
- Products and Categories: A product belongs to only one category, but a category can have many products. This is represented by the foreign key category_id in the products table. [One-to-Many Relationships]
- Channels and Customers: A customer can register through multiple channels, and a channel can have many registered customers. [Many-to-Many Relationships]

List of tables

- Customers
    - It contains information about customers.
    - It is used to store and manage customer details for communication, analysis, and order tracking.
- Orders
    - It contains details of customer orders.
    - It is used to track order details, including who placed the order, where it should be shipped, and the overall cost.
- order_items
    - It details the products included in each order.
    - To manage and track the specific products within an order, allowing for detailed analysis.
- Products
    - Contains information about the products available for sale.
- Categories
    - Contains information about product categories.
- Channels
    - Details the channels through which customers are acquired.
    - Example:

        - Social
        - Referral
        - Email
        - Paid search
        - Etc.

Description About Attributes

Customers

- customer_id
    - Unique identifier for each customer.
    - Type: int
- Email
    - The email address of the customer.
    - Type: String
- full_name
    - The full name of the customer.
    - Type: String
- Address
    - The customer's street address
    - Type: String
- City
    - The city where the customer resides.
    - Type: String
- Region
    - The region, state, or province where the customer resides
    - Type: String
- postal_code
    - The postal or ZIP code for the customer's address
    - Type: String

- Country
    - The country where the customer resides
    - Type: String
- Phone
    - The customer's phone number, used for communication and verification
    - Type: String
- registration_date
    - The date when the customer registered.
    - Type: Date
- channel_id
    - A reference to the channel through which the customer was acquired.
    - Type: Int (Foreign Key)
- first_order_id
    - The unique identifier of the customer's first order.
    - Type: int
- first_order_date
    - The date when the customer placed their first order.
    - Type: date
- last_order_id
    - The unique identifier of the customer's most recent order.
    - Type: int
- last_order_date
    - The date when the customer placed their most recent order.
    - Type: date

Orders

- order_id
    - A unique identifier for each order.
    - Type: int

- customer_id
    - References the unique identifier of the customer
    - Type: int (foreign key)
- order_date
    - The date when the order was placed.
    - Type: date
- total_amount
    - The total monetary value of the order.
    - Tyoe: decimal
- ship_name
    - The name of the person or entity to whom the order is being shipped.
    - Type: string
- ship_address
    - The street address where the order is to be delivered
    - Type: String
- ship_city
    - The city where the order is to be delivered
    - Type: string
- ship_region
    - The region, state, or province where the order is to be delivered
    - Type: String
- ship_postal_code
    - The postal or ZIP code for the shipping address
    - Type: string
- ship_country
    - The country where the order is to be delivered.
    - Type: string
- shipped_date

    - The date when the order was shipped.
    - Type: date

products

- product_id
    - A unique identifier for each product.
    - Type: int
- product_name
    - The name of the product.
    - Type: string
- category_id
    - References the unique identifier of the category.
    - Type: int
- unit_price
    - The price of a single unit of the product.
    - Type: decimal
- Discontinued
    - Indicates whether the product is still available for sale.
    - Type: tinyint

order_items

- order_id
    - A unique identifier for each order
    - Type: int
- product_id
    - References the unique identifier of the product being ordered.
    - Type: int
- unit_price
    - The price per unit of the product at the time of the order
    - Type: decimal

- quantity
    - The number of units of the product being ordered.
    - Type: int
- Discount
    - The discount applied to the product.
    - Type: decimal

categories

- category_id
    - A unique identifier for each category.
    - Type: int
- category_name
    - The name of the category.
    - Type: string
- Description
    - A detailed description of the category.
    - Type: text

channels

- Id
    - A unique identifier for each channel.
    - Type: int
- channel_name
    - The name of the channel.
    - Type: string

Questions

1. How many customers registered in the first six months of 2017? Name the column registration_count.

Select count(customer_id) as registration_count from customers where year(registration_date)=2017 and month(registration_date)<=6;

	registration_count
▶	34

2. Show the number of registrations in the current week. Name the column registrations_current_week.

Select count(customer_id) as registrations_current_week

From customers where week(registration_date)= week (current_date);

	registrations_current_week
▶	0

3. Create a report containing the 2017 monthly registration counts. Show the registration_month and registration_count columns. Order the results by month.

Ans:

Select month(registration_date) as registration_month ,

count(customer_id) as registration_count

From customers where year(registration_date)=2017 group by month(registration_date) order by month(registration_date);

	registration_month	registration_count
▶	1	16
	2	15
	3	2
	5	1

4. Find the registration count for each month in each year. Show the following columns: registration_year, registration_month, and registration_count. Order the results by year and month.

Select year(registration_date) as registration_year ,month(registration_date) as registration_month , count(*)as registration_count From customers group by year(registration_date), month(registration_date)order by year(registration_date),month(registration_date);

registration_year	registration_month	registration_count
2016	12	6
2017	1	16
2017	2	15
2017	3	2
2017	5	1
2018	12	1
2024	2	3
2024	3	25
2024	4	14
2024	5	1

5. Write an SQL query to find the number of customer registrations per year for each channel.

Select channel_id ,year(registration_date),count(customer_id) from customers Group by channel_id ,year(registration_date) order by channel_id ,year(registration_date);

channel_id	year(registration_date)	count(customer_id
1	2016	4
1	2017	24
1	2018	1
1	2024	32
2	2016	2
2	2017	4
2	2024	6
3	2017	4

6. Write an SQL query to find the number of customer registrations per year for organic search channel.

Select year(registration_date),count(customer_id)

from customers c join channels ch on c.channel_id = ch.id

where ch.channel_name = 'Organic Search 'group by year(registration_date);

year(registration_date)	count(customer_id)

7 Create a report to show the weekly counts of registration in 2017, based on the customer country. Show the following columns: registration_week, country, and registration_count. Order the results by week.

```
Select country, week(registration_date) as registration_week ,

count(customer_id)as registration_count

from customers where year(registration_date) =2017

group by country, week(registration_date)

order by week(registration_date)  ;
```

Result Grid | Filter Rows: | Expor

country	registration_week	registration_count
Germany	1	1
Mexico	1	2
France	1	2
Italy	1	1
Germany	2	2
Spain	2	1
France	2	1
Venezuela	2	1

8 . Among customers registered in 2017, show how many made at least one purchase (name the column customers_with_purchase) and the number of all the customers registered in 2017 (name the column all_customers).

```
Select count(customer_id) as all_customers,

Count(first_order) as customers_with_purchase  FROM customers where
year(registration_date) = 2017 ;
```

all_customers	customers_with_purchase
34	25

9 Find the lifetime conversion rate among customers who registered in 2017. Show the result in a column named conversion_rate. Round the result to four decimal places.

Select round(count(first_order_id)/count(customer_id),4) as conversion_rate from customers where year(registration_date)=2017;

Result Grid

conversion_rate
0.7353

10 Find the conversion rate for each customer channel. Show the channel_name and conversion_rate columns. Display the conversion rates as percentages rounded to two decimal places.

Select ch.channel_name as channel_name ,
round(count(first_order_id)/count(customer_id)*100,2) as conversion_rate
from customers c join channels ch on c.channel_id = ch.id group by ch.channel_name ;

Result Grid Filter Rows:

channel_name	conversion_rate
Organic Search	54.10
Referral	62.50
Paid Search	100.00
Direct	58.33
Social	100.00

11 Create a report showing conversion rates in monthly basis. Display the conversion rates as ratios, rounded to three decimal places. Show the following columns: year, month, and conversion_rate. Order the results by year and month.

Select year(registration_date) as year, month(registration_date) as month
,round(count(first_order_id)/count(customer_id),3) as conversion_rate from
customers c join channels ch on c.channel_id = ch.id group by year(registration_date),
month(registration_date) order by year(registration_date), month(registration_date);

year	month	conversion_rate
2016	12	1.000
2017	1	0.813
2017	2	0.667
2017	3	1.000
2017	5	0.000
2018	12	0.000
2024	2	1.000
2024	3	0.400

12 Create a report containing the conversion rates for weekly registration in each registration channel, based on customers registered in 2017. Show the following columns: week, channel_name, and conversion_rate. Format the conversion rates as percentages, rounded to a single decimal place. Order the results by week and channel name.

```
Select week(registration_date) as week,

ch.channel_name as channel_name ,

round(count(first_order_id)/count(customer_id *100),1) as conversion_rate
from customers c join channels ch on  c.channel_id = ch.id

where year(registration_date)=2017

group by week(registration_date),

ch.channel_name

order by week(registration_date), ch.channel_name;
```

week	channel_name	conversion_rate
1	Direct	1.0
1	Organic Search	0.8
1	Paid Search	1.0
2	Direct	1.0
2	Organic Search	0.7
2	Referral	1.0
3	Organic Search	0.7
4	Organic Search	1.0

13 . Show customers' emails and interval between their first purchase and the date of registration. Name the column difference.

Select email , datediff(first_order_date , registration_date) as difference from customers where first_order_date is not null ;

email	difference
michael.holz@course.vertabelo.com	14
ana.trujillo@course.vertabelo.com	10
antonio.moreno@course.vertabelo.com	19
hanna.moos@course.vertabelo.com	2
frederique.citeaux@course.vertabelo.com	3
martin.sommer@course.vertabelo.com	4
patricio.simpson@course.vertabelo.com	8
francisco.chang@course.vertabelo.com	10

14 Find the average time from registration to first order for each channel. Show two columns: channel_name and avg_days_to_first_order.

Select ch.channel_name ,

avg(datediff(first_order_date , registration_date)) as avg_days_to_first_order

from customers c join channels ch on c.channel_id = ch.id

where first_order_date is not null group by ch.channel_name;

Result Grid Filter Rows:

channel_name	avg_days_to_first_order
Organic Search	12.1818
Paid Search	12.5000
Referral	9.0000
Direct	9.8571
Social	16.0000

15 Calculate the average number of days that passed between registration and first order in quarterly registration basis. Show the following columns: year, quarter, and avg_days_to_first_order. Order the results by year and quarter.

```
select
        year(registration_date) ,
        case
                when month(registration_date) in (1,2,3) then 'first_quarter'
                when month(registration_date) in (4,5,6) then 'second_quarter'
                when month(registration_date) in (7,8,9) then 'third_quarter'
                when month(registration_date) in (10,11,12) then 'fourth_quarter'
        end as quarterrly_registaration,
        avg(datediff(first_order_date , registration_date)) as avg_days_to_first_order
from customers
where
        first_order_date is not null
group by
        year(registration_date) , quarterrly_registaration
order by
        year(registration_date) , quarterrly_registaration ;
```

another approach :

```
SELECT
   YEAR(registration_date) AS year,
   QUARTER(registration_date) AS quarter,
   AVG(DATEDIFF(first_order_date, registration_date)) AS avg_days_to_first_order
FROM
```

```
   customers
WHERE
   first_order_date IS NOT NULL
GROUP BY
   YEAR(registration_date), QUARTER(registration_date)
ORDER BY
   YEAR(registration_date), QUARTER(registration_date);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Conten

year(registration_date)	quarterrly_registaration	avg_days_to_first_order
2016	fourth_quarter	15.5000
2017	first_quarter	11.5200
2024	first_quarter	11.1538
2024	second_quarter	7.7500

16 Create a report of the average time to first order for weekly registration basis from 2017 in each registration channel. Show the following columns: week, channel_name, and avg_days_to_first_order. Order the results by the week.

```
Select
        week(registration_date),ch.channel_name ,
        avg(datediff(first_order_date , registration_date)) as avg_days_to_first_order
from
        customers c join  channels ch on c.channel_id = ch.id
where
        year(registration_date)=2017  and first_order_date  is not null
group by
```

```
        week(registration_date), ch.channel_name
order by
        week(registration_date);
```

Result Grid | Filter Rows: | Export: | W

week(registration_date)	channel_name	avg_days_to_first_order
1	Direct	3.0000
1	Organic Search	14.0000
1	Paid Search	10.0000
2	Direct	4.0000
2	Organic Search	22.5000
2	Referral	2.0000
3	Organic Search	14.0000
4	Organic Search	7.0000

17. Find all customers who placed their first order within one month from registration, and their last order within three months from registration – let's see who's stopped ordering. For each customer show these columns: email, full_name, first_order_date, last_order_date.

```
Select
        email, full_name,
        first_order_date,
  last_order_date
from
        customers
  where
first_order_date is not null
and DATEDIFF(first_order_date, registration_date) between 1 and  30
and DATEDIFF(last_order_date, registration_date) < 90;
```

email	full_name	first_order_date	last_order_dat
ricky.bosse@course.vertabelo.com	Ricky Bosse	2024-03-03	2024-03-03
alyce.beasley@course.vertabelo.com	Alyce Beasley	2024-03-14	2024-05-17
katrina.lambright@course.vertabelo.com	Katrina Lambright	2024-02-25	2024-02-25
michael.montes@course.vertabelo.com	Michael Montes	2024-04-16	2024-04-16
edgar.allen@course.vertabelo.com	Edgar Allen	2024-04-03	2024-06-17
kevin.oconnell@course.vertabelo.com	Kevin Oconnell	2024-04-17	2024-06-28
thomas.helmick@course.vertabelo.com	Thomas Helmick	2024-03-27	2024-06-17
joseph.ramirez@course.vertabelo.com	Joseph Ramirez	2024-04-16	2024-06-23

18 . Our e-store has used three versions of the registration form:

'ver1' – introduced when the e-store started.

'ver2' – introduced on Mar 14, 2017.

'ver3' – introduced on Jan 1, 2018.

```
SELECT
  registration_date,
  CASE
    WHEN registration_date < '2017-03-14' THEN 'ver1'
    WHEN registration_date >= '2017-03-14' AND registration_date < '2018-01-01' THEN 'ver2'
    WHEN registration_date >= '2018-01-01' THEN 'ver3'
  END AS form_version
FROM customers;
```

Result Grid | Filter Rows:

registration_date	form_version
2024-05-09	ver3
2017-01-17	ver1
2017-05-24	ver2
2024-03-08	ver3
2017-03-17	ver2
2017-01-03	ver1
2017-01-04	ver1
2017-01-05	ver1

Result 1

19 Show two metrics in two different columns:

order_on_registration_date – the number of people who made their first order within one day from their registration date.

order_after_registration_date – the number of people who made their first order after their registration date.

Ans:

```
Select

count(case first_order_date  is not null and datediff(first_order_date, registration_date)=0
then customer_id end) as order_on_registration_date,

count(case first_order_date  is not null and datediff(first_order_date, registration_date)>0
then customer_id end) as order_after_registration_date

from customers;
```

order_on_registration_date	order_after_registration_date
2	46

20 Create a conversion chart for monthly registration. Show the following columns:

year

month

registered_count

no_sale

three_days – the number of customers who made a purchase within 3 days from registration.

first_week – the number of customers who made a purchase during the first week but not within the first three days.

after_first_week – the number of customers who made a purchase after the 7th day.

Order the results by year and month.

Ans:

```
Select
        year(registration_date ) as year,
        month(registration_date) as month,
        count(customer_id ),
        count( case  when first_order_date is null then customer_id end )as  'no sales ',
        count( case  when first_order_date is not null and datediff( first_order_date
,registration_date) between 0 and 2 then customer_id end ) as  'three_days' ,
        count(case  when first_order_date is not null and datediff(first_order_date
,registration_date) between 3 and 6 then customer_id end )as  'one week',
        count(case  when first_order_date is not null and datediff(first_order_date
,registration_date ) >6  then customer_id  end )as 'after_first_week '
from
        customers
group by
year, month
order by year, month;
```

year	month	count(customer_id)	no sales	three_days	one week	after_first_week
2016	12	6	0	0	0	6
2017	1	16	3	1	2	10
2017	2	15	5	2	1	7
2017	3	2	0	0	0	2
2017	5	1	1	0	0	0
2018	12	1	1	0	0	0
2024	2	3	0	0	1	2